

1.1 QUBIT (Quantum Bit)

A qubit, short form of "quantum bit," is a fundamental unit of information in quantum computing. It is the quantum analog of a classical bit, which is represented as 0 or 1 in classical computing.

A qubit is a two-level quantum system where the two basis qubit states are usually written as $|0\rangle$ and $|1\rangle$. A qubit can be in state $|0\rangle$ or $|1\rangle$ or in a linear combination of both states.

When we discuss qubit basis states, we essentially alluding to the z-basis states as the fundamental computational states.

Properties of qubits

1. Qubits make use of discrete energy state particles such as electrons and photons.
2. Qubits exists in two quantum state $|0\rangle$ and $|1\rangle$ or in a linear combination of both states. This is known as superposition.
3. Unlike classical bits, qubit can work with the overlap of both 0 & 1 states.
4. When the qubit is measured, it collapses to one of the two basis states $|0\rangle$ or $|1\rangle$
5. Quantum entanglement and quantum tunnelling are two exclusive properties of qubit.
6. State of the qubits is represented using Bloch sphere.

Example :

Suppose there are two basis Qubit states $|0\rangle$ and $|1\rangle$

These states are such that

$$\langle 0|0\rangle = \langle 1|1\rangle = 1 \quad (\text{Normalized basis})$$

$$\langle 0|1\rangle = \langle 1|0\rangle = 0 \quad (\text{orthogonal})$$

Any linear combination of these two basis states will also be a basis states.

$$|\psi\rangle = 0.5|0\rangle + 0.2i|1\rangle$$

Or in general

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Important Note: Where a and b are the complex probability amplitude for respective basis states. The condition on a and b means that $|\Psi\rangle$ should be normalised.

$$|a|^2 + |b|^2 = 1$$

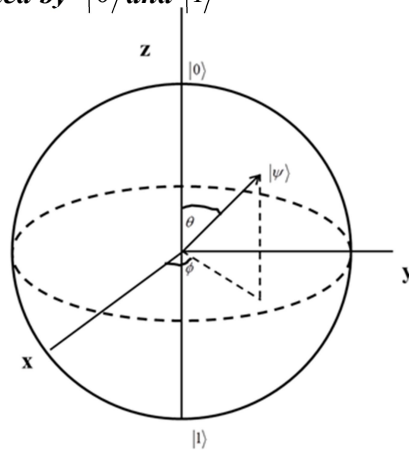
Example: suppose basis Qubit states are represented by $|0\rangle$ and $|1\rangle$

Then its linear combination can also be written as

$$|\psi\rangle = |\theta, \phi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right)|1\rangle$$

For $\theta=0$ and $\phi=0$

$$|\psi\rangle = |0, 0\rangle = \cos(0)|0\rangle + e^{i0} \sin(0)|1\rangle = |0\rangle$$



For $\theta=\frac{\pi}{2}$ and $\phi=0$

$$|\psi\rangle = \left|\frac{\pi}{2}, 0\right\rangle = \cos\left(\frac{\pi}{4}\right)|0\rangle + e^{i0} \sin\left(\frac{\pi}{4}\right)|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

For $\theta=\pi$ and $\phi=0$

$$|\psi\rangle = |\pi, 0\rangle = \cos\left(\frac{\pi}{2}\right)|0\rangle + e^{i0} \sin\left(\frac{\pi}{2}\right)|1\rangle = |1\rangle$$

1.2 Qudit:

A Qudit or a d-dimensional quantum system can be represented as d-dimensional ket vector $|\Psi\rangle$

$$|\Psi\rangle = \sum_{i=0}^{d-1} \alpha_i |\phi_i\rangle$$

provided

$$\sum_{i=0}^{d-1} |\alpha_i|^2 = 1$$

The condition on the coefficients α_i means that $|\Psi\rangle$ is a vector of length of 1.

1.3 STANDARD BASIS:

Consider a 2 dimensional orthonormal complex vector space. The standard basis also sometimes known as computational basis $\mathcal{S} = \{|0\rangle, |1\rangle\}$, is an orthonormal basis for this vector space, where the basis vectors are

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

1.4 HADAMARD BASIS:

The Hadamard basis is an orthonormal basis $\mathcal{H} = \{|0\rangle, |1\rangle\}$, is an orthonormal basis for this vector space, where the basis vectors are

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Let us check this is an orthonormal basis

$$\langle + | - \rangle = \langle - | + \rangle = 0, \text{ \& } \langle + | + \rangle = \langle - | - \rangle = 1$$

Example: Express $|1\rangle$ in the Hadamard basis. That is, find coefficients α and β such that $|1\rangle = \alpha|+\rangle + \beta|-\rangle$.

Since we know that standard basis or computational basis is defined as

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Therefore Hadamard basis can be written as

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$|1\rangle = \alpha|+\rangle + \beta|-\rangle = \alpha \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \beta \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

$$\Rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \alpha \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \beta \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} \alpha + \beta \\ \alpha - \beta \end{bmatrix}$$

$$\Rightarrow \alpha + \beta = 0$$

$$\Rightarrow \alpha - \beta = \sqrt{2}$$

$$\Rightarrow \alpha = -\beta = \frac{1}{\sqrt{2}}$$

$$\Rightarrow |1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$$

1.5 MULTIPLE QUBIT

Classically if we have two bits, we write as “00”, “01” and so on.

How can we write two Qubits?

For example: suppose we want to write $|00\rangle$, $|11\rangle$

Since we know that standard basis or computational basis is defined as

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Before solving above problem we should learn **Kronecker or Tensor product of two Matrices**

$$A=|A\rangle=\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \text{ and } B=|B\rangle=\begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix}$$

Therefor tensor product of two matrices can be written as

$$A \otimes B = \begin{bmatrix} 1 \times \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix} & 2 \times \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix} \\ 3 \times \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix} & 4 \times \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix} \end{bmatrix}$$

$$\Rightarrow A \otimes B = \begin{bmatrix} 0 & 5 & 0 & 10 \\ 6 & 7 & 8 & 14 \\ 0 & 15 & 0 & 20 \\ 18 & 21 & 24 & 28 \end{bmatrix}$$

Let us do the example for how to write the multiple Qubit?

$$A = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$B = |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$A \otimes A = |00\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot B = \begin{bmatrix} 1 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ 0 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$B \otimes B = |11\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \cdot B = \begin{bmatrix} 0 \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ 1 \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Example: Consider two qubits A and B, in the two qubit state known as the EPR pair one can label the joint state as AB?

$$|EPR\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB}) = \frac{1}{\sqrt{2}}\left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}\right) = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Where EPR stands for Einstein, Podolsky and Rosen, (we will study later).

Which is equal to the superposition between the vectors $|00\rangle_{AB}$ and $|11\rangle_{AB}$.

The length of this vector is calculated using inner product.

$$\langle EPR|EPR\rangle_{AB} = \frac{1}{\sqrt{2}}(\langle 00|_{AB} + \langle 11|_{AB}) \times \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB})$$

$$\Rightarrow \langle EPR|EPR\rangle_{AB} = \frac{1}{2}([1 \ 0 \ 0 \ 0] + [0 \ 0 \ 0 \ 1]) \times \left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right)$$

$$\Rightarrow \langle EPR|EPR\rangle_{AB} = \frac{1}{2} \left([1 \ 0 \ 0 \ 0] \times \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + [1 \ 0 \ 0 \ 0] \times \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} + [0 \ 0 \ 0 \ 1] \times \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + [0 \ 0 \ 0 \ 1] \times \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right)$$

$$\Rightarrow \langle EPR|EPR\rangle_{AB} = \frac{1}{2}(\langle 00|00\rangle_{AB} + \langle 00|11\rangle_{AB} + \langle 11|00\rangle_{AB} + \langle 11|11\rangle_{AB})$$

$$\Rightarrow \langle EPR|EPR\rangle_{AB} = \frac{1}{2}(1 + 0 + 0 + 1) = 1$$

1.6 How to combine Qubits?

Let us consider we have two Qubits A and B. We describe the state A as $|\Psi\rangle_A$ and state B as $|\Phi\rangle_B$.

$$|\Psi\rangle_A = \alpha_A |0\rangle_A + \beta_A |1\rangle_A$$

$$|\Phi\rangle_B = \alpha_B |0\rangle_B + \beta_B |1\rangle_B$$

Tensor product of two vectors

$$|\Psi\rangle_{AB} = |\Psi\rangle_A \otimes |\Phi\rangle_B = \begin{bmatrix} \alpha_A \alpha_B \\ \alpha_A \beta_B \\ \beta_A \alpha_B \\ \beta_A \beta_B \end{bmatrix}$$

1.7 Operators

Operators are actions on quantum states that transform them.

For a quantum state represented by ket $|\psi\rangle$, an operator $\hat{A}$ can act on it and transform it into another quantum state represented by ket $|\phi\rangle$.

$$\hat{A}|\Psi\rangle = |\phi\rangle$$

Example: Suppose

$$|\Psi\rangle = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\text{Suppose an operator } \hat{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\hat{X}|\Psi\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |\phi\rangle$$

1.8 Identity Operator

The identity operator is an operator which, operating on a function, leaves the function unchanged $I|\Psi\rangle = |\Psi\rangle$ where $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$|\Psi\rangle = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Suppose an operator I operates on $|\Psi\rangle$

$$I|\Psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |\Psi\rangle$$

$$I|\Psi\rangle = |\Psi\rangle = |0\rangle$$

1.9 Conjugate matrices:

If the elements in a matrix A are complex numbers, then the matrix obtained by the corresponding conjugate complex elements is called the conjugate of A and is denoted by A^* .

Suppose $A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$ therefore the complex conjugate of the matrix A can be written as $A^* = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$

1.10 Transpose Matrices

If columns and rows of a matrix A are interchanged then the resultant matrix is transpose of A and represented as A^T .

Suppose $A = \begin{bmatrix} 0 & 1 \\ -i & 0 \end{bmatrix}$ therefore the Transpose of the matrix A can be written as $A^T = \begin{bmatrix} 0 & -i \\ 1 & 0 \end{bmatrix}$

1.11 Hermitian Matrices

The transpose of complex conjugate of a matrix is known as Hermitian operator and the resultant matrix is known as Hermitian matrix if $A = A^{*T} = A^\dagger$. It is represented by $A^\dagger$

Suppose $A = \begin{bmatrix} 8 & 1+i \\ 1-i & 5 \end{bmatrix}$ therefore the complex conjugate of matrix A is $A^* = \begin{bmatrix} 8 & 1-i \\ 1+i & 5 \end{bmatrix}$ and Transpose of the matrix A^* can be written as $A^{*T} = \begin{bmatrix} 8 & 1+i \\ 1-i & 5 \end{bmatrix}$

Therefore A is called the Hermitian matrix if $A = A^{*T} = A^\dagger$

1.12 Unitary Matrix

A matrix is said to be Unitary if

$$AA^\dagger = I$$

1.13 Pauli Matrices

$$\sigma_x = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \sigma_z = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\sigma_y = Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

1.14 Properties of Pauli's Matrices

(a) Square Pauli matrices gives identity matrix I

$$X^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$X^2 = I$$

Similarly

$$Y^2 = I$$

$$Z^2 = I$$

(b) Pauli matrices are unitary matrix

$$XX^\dagger = I$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Similarly

$$YY^\dagger = I$$

$$ZZ^\dagger = I$$

(c) Pauli matrices are Hermitian:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$X^\dagger = X^{*T} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$X = X^\dagger \text{ Similarly}$$

$$Y = Y^\dagger$$

$$Z = Z^\dagger$$

1.15 Quantum Gates:

A quantum gate is a very simple computing device that performs quantum operation on qubits. Quantum gates are one of the essential parts of a quantum computer and are the building blocks of all quantum algorithms.

Quantum gates are mathematically represented as transformation matrices which operate on inputs to give outputs.

There are different types of quantum gates. Single-qubit gates and multiple qubit gates. These gates can flip a qubit from 0 to 1 as well as allowing superposition states to be created.

1.16 Single-Qubit Gates:

As we have studied earlier Single Qubits are $|0\rangle$ and $|1\rangle$.

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Single Qubit Gates are X- gate, Y- gate, Z- gate, H –gate, S- gate, T- gate.

(i) X – Gate or Quantum Not Gate

X-gate is the single qubit input gate and is also called as Pauli X – Gate or quantum NOT Gate.

The Matrix form of X is given by $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Operation of X- Gate on $|0\rangle, |1\rangle$, and $|\Psi\rangle$ can be evaluated as following.

$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

$$\Rightarrow X|0\rangle = |1\rangle$$

$$\text{Similarly } X|1\rangle = |0\rangle$$

Since X inverts each input it is also called as **bit-flip gate**.

$$\text{If } |\Psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \text{ therefore } X|\Psi\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix} = \alpha|1\rangle + \beta|0\rangle$$

$$\alpha|0\rangle + \beta|1\rangle \xrightarrow{\text{X}} \alpha|1\rangle + \beta|0\rangle$$

X-Gate Truth Table

X-Gate Truth Table	
Input	Output
$ 0\rangle$	$ 1\rangle$
$ 1\rangle$	$ 0\rangle$
$\alpha 0\rangle + \beta 1\rangle$	$\alpha 1\rangle + \beta 0\rangle$

(ii) Y – Gate or Pauli Y – gate.

$$\text{The matrix form of Y gate is } Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

Operations of Y gate on $|0\rangle$, and $|1\rangle$ provides following output

$$Y|0\rangle = i|1\rangle$$

$$\Rightarrow Y|0\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} = i \begin{bmatrix} 0 \\ 1 \end{bmatrix} = i|1\rangle$$

$$\text{Similarly } Y|1\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -i \\ 0 \end{bmatrix} = -i \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Y|1\rangle = -i|0\rangle$$

If a superposed Qubit goes through Y-Gate

$$\text{If } |\Psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \text{ therefore } Y|\Psi\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} -i\beta \\ i\alpha \end{bmatrix} = -i\beta|0\rangle + i\alpha|1\rangle$$

$$\alpha|0\rangle + \beta|1\rangle \xrightarrow{\text{Y}} -i\beta|0\rangle + i\alpha|1\rangle$$

Y-Gate Truth Table

Y-Gate Truth Table	
Input	Output
$ 0\rangle$	$i 1\rangle$
$ 1\rangle$	$-i 0\rangle$
$\alpha 0\rangle + \beta 1\rangle$	$i\alpha 1\rangle - i\beta 0\rangle$

(iii) Z – Gate : Z-gate is the single qubit input gate. This is also called as Pauli Z – gate.

The matrix form of Z gate is $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Action of Z gate on inputs: When Z operates on $|0\rangle$ and $|1\rangle$ the phase will change. Hence this is also called as phase-flip gate

$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

$$Z|0\rangle = |0\rangle$$

$$Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -|1\rangle$$

$$Z|1\rangle = -|1\rangle$$

$$Z|\Psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ -\beta \end{bmatrix} = \alpha|0\rangle - \beta|1\rangle$$

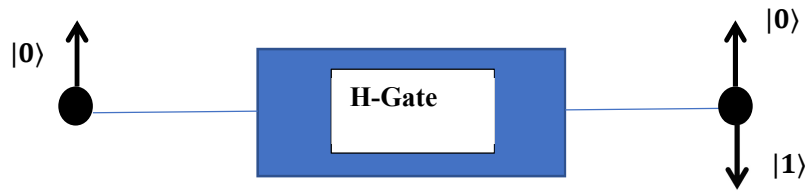
$$Z|\Psi\rangle = \alpha|0\rangle - \beta|1\rangle$$

$$\alpha|0\rangle + \beta|1\rangle \xrightarrow{\text{Z}} \alpha|0\rangle - \beta|1\rangle$$

Z-Gate Truth Table

Z-Gate Truth Table	
Input	Output
$ 0\rangle$	$ 0\rangle$
$ 1\rangle$	$- 1\rangle$
$\alpha 0\rangle + \beta 1\rangle$	$\alpha 0\rangle - \beta 1\rangle$

(iv) **Hadamard Gate (H-gate)** – It is a single qubit gate and is also gate to superposition. H gate acts on single qubit input and produce superposition state output. Matrix form of H gate and its symbol is $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

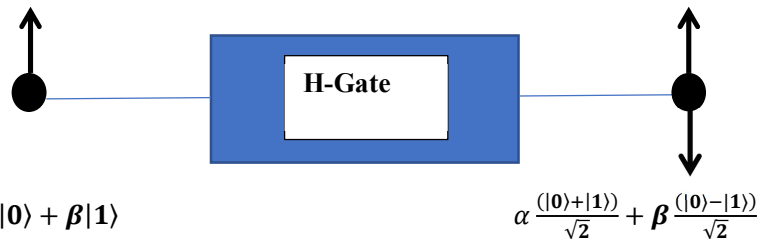


$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$$H|\Psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} \alpha + \beta \\ \alpha - \beta \end{bmatrix} = \frac{(\alpha + \beta)}{\sqrt{2}} |0\rangle + \frac{(\alpha - \beta)}{\sqrt{2}} |1\rangle$$

$$H|\Psi\rangle = \alpha \frac{(|0\rangle + |1\rangle)}{\sqrt{2}} + \beta \frac{(|0\rangle - |1\rangle)}{\sqrt{2}}$$



H-Gate Truth Table

H-Gate Truth Table	
Input	Output
$ 0\rangle$	$\frac{(0\rangle + 1\rangle)}{\sqrt{2}}$
$ 1\rangle$	$\frac{(0\rangle - 1\rangle)}{\sqrt{2}}$
$\alpha 0\rangle + \beta 1\rangle$	$\alpha \frac{(0\rangle + 1\rangle)}{\sqrt{2}} + \beta \frac{(0\rangle - 1\rangle)}{\sqrt{2}}$

Note: Difference between X, Y, Z and H gates is that in X, Y and Z gates, output is in single state whereas in H gate output is superposed state.

Compton Effect: particle nature of light

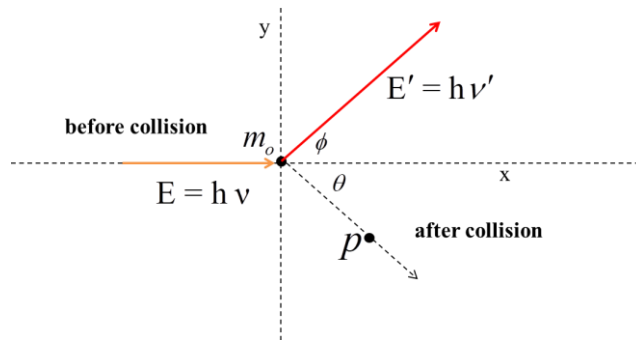


Fig.1

Let us consider a photon of energy $E = h\nu$ incident on an electron having rest mass m_0 and which is initially at rest. After the collision the photon having energy $E' = h\nu'$ is scattered at an angle ϕ and an electron moves with momentum p and making an angle θ from the x axis.

Now according to energy conservation principle

Loss in photon energy = gain in kinetic energy of the electron

$$h\nu - h\nu' = KE \dots\dots\dots(1)$$

Applying momentum conservation along x and y directions

$$\frac{h\nu}{c} = \frac{h\nu'}{c} \cos \phi + p \cos \theta$$

$$p \cos \theta = h\nu - h\nu' \cos \phi \dots\dots\dots(2)$$

$$p \sin \theta = h\nu' \sin \phi \dots\dots\dots(3)$$

Now squaring and adding equations (2) and (3)

$$p^2 c^2 = (h\nu)^2 + (h\nu')^2 - 2h^2 \nu \nu' \cos \phi \dots\dots\dots(4)$$

Now since the total energy of the particle is defined as

$$E = KE + m_0 c^2 \dots\dots\dots(5)$$

Relation between total energy, momentum and rest energy of the particle

$$E^2 = p^2 c^2 + m_0^2 c^4 \dots\dots\dots(6)$$

Substituting the value of equation (1) in equation (5)

$$E = (h\nu - h\nu') + m_0 c^2 \dots\dots\dots(7)$$

Now take the square of equation (7) and substituting in (6), we have

$$\begin{aligned} \{(hv - hv') + m_0c^2\}^2 &= p^2c^2 + m_0^2c^4 \\ (hv)^2 + (hv')^2 - 2h^2vv' + m_0^2c^4 + 2hm_0c^2(v - v') &= p^2c^2 + m_0^2c^4 \\ (hv)^2 + (hv')^2 - 2h^2vv' + 2hm_0c^2(v - v') &= p^2c^2 \dots\dots\dots(8) \end{aligned}$$

Now comparing equations (4) and (8) we have

$$\begin{aligned} (hv)^2 + (hv')^2 - 2h^2vv' + 2hm_0c^2(v - v') &= (hv)^2 + (hv')^2 - 2h^2vv'\cos\phi \\ 2hm_0c^2(v - v') &= 2h^2vv' - 2h^2vv'\cos\phi \\ m_0c^2(v - v') &= hvv'(1 - \cos\phi) \\ \frac{(v-v')}{vv'} &= \frac{h}{m_0c^2}(1 - \cos\phi) \dots\dots\dots(9) \end{aligned}$$

Since $c = v\lambda$

$$v = \frac{c}{\lambda}$$

$$\frac{1}{v} = \frac{\lambda}{c}$$

$$\Delta\lambda = (\lambda' - \lambda) = \frac{h}{m_0c}(1 - \cos\phi)$$

Where $\lambda_c = \frac{h}{m_0c}$ is known as Compton wavelength.

$$\lambda_c = \frac{6.626 \times 10^{-34}}{9.1 \times 10^{-31} \times 3.0 \times 10^8} = 2.426 \times 10^{-12} \text{meter} = 2.426 \text{ pm}$$

de Broglie Wave phase and group velocities

1. **de Broglie wave:** The de Broglie wavelength λ for a particle that has certain mass (rest mass m_0) is related to its momentum p as

$$\lambda = \frac{h}{p} = \frac{h}{\gamma m_0 v} \dots \dots \dots (1)$$

Where v is the velocity of the particle, and $\gamma = \frac{1}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}}$

If $v \ll c$ (velocity of light) then

$$\lambda = \frac{h}{p} = \frac{h}{m_0 v} \dots \dots \dots (2)$$

Since the de Broglie wave is associated with a moving body one can expect that the wave should have the same velocity as the particle.

Let us check?

If we assume v_p the velocity of the de Broglie wave associated with the particle and v is the velocity of the particle.

2. **Phase Velocity:** The velocity v_p with which the de Broglie wave travels is called phase velocity and is defined by following equation (3).

The according to wave theory we can write

$$v_p = v\lambda \dots \dots \dots (3)$$

Where v is the frequency of the de Broglie wave and λ is the wavelength of the de Broglie wave.

So according to de Broglie the wavelength λ can be written as

$$\lambda = \frac{h}{p} = \frac{h}{\gamma m_0 v} \dots \dots \dots (4)$$

We also know that from mass energy relation

$$E = hv = \gamma m_0 c^2$$

$$v = \frac{\gamma m_0 c^2}{h} \dots \dots \dots (5)$$

Now from equation (3) we can find out phase velocity of the de Broglie wave using equations (4) and (5).

$$v_p = \frac{\gamma m_0 c^2}{h} \times \frac{h}{\gamma m_0 v}$$

$$v_p = \frac{c^2}{v} \dots \dots \dots (6)$$

From equation (6) we infer that velocity of the de Broglie wave is faster than the velocity of particle as $v \ll c$. Further since $v \ll c$ so $v_p \gg c$, which means de Broglie wave travels much faster than the velocity of light. This is wrong result as phase velocity cannot be greater the light velocity. ***In order to understand this discrepancy we must look the distinction between phase and group velocity.***

3. **Group velocity(v_g):** de Broglie wave associated with a moving particle consists of a group of wave rather a single wave and velocity with which this group of wave packet travel is called the group velocity.

$$\text{Group velocity } v_g = \frac{d\omega}{dk} \dots \dots \dots (7)$$

Let us check whether the group of wave associated with the moving particle travels with the same velocity as particle moves?

Since we know that the angular frequency ω is defined as

$$\omega = 2\pi\nu \dots \dots \dots (8)$$

Using equation (5) $\omega = \frac{2\pi\gamma m_0 c^2}{h} \dots \dots \dots (9)$

$$\omega = \frac{2\pi m_0 c^2}{h \left(1 - \frac{v^2}{c^2}\right)^{\frac{1}{2}}} \dots \dots \dots (10)$$

Also wave number

$$k = \frac{2\pi}{\lambda} \dots\dots\dots(11)$$

Using equation (4) we can get

$$k = \frac{2\pi m_0 v}{h \left(1 - \frac{v^2}{c^2}\right)^{\frac{1}{2}}} \dots\dots\dots(12)$$

From equation (7) group velocity of the group of waves can be written as

$$v_g = \frac{d\omega}{dk} = \frac{d\omega/dv}{dk/dv} \dots\dots\dots(13)$$

After putting the values of $\frac{d\omega}{dv}$ and $\frac{dk}{dv}$ after differentiating equations (10) and (12)

With respect to v , we get

$$v_g = v \dots\dots\dots(14)$$

Thus the de Broglie wave group associated with the moving particles travel with the same velocity as the particle.

Heisenberg Uncertainty Principle:

It is impossible to know both the exact position and exact momentum of an object (particle) at the same time.

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

Problem 1: Uncertainty in position of electron $\Delta x = 50 \text{ pm}$, find out the uncertainty in the momentum of the electron?

Solution: Given that uncertainty in the position $\Delta x = 50 \text{ pm}$

$$\Delta x = 50 \times 10^{-12} \text{ m}$$

Now from Heisenberg uncertainty principle

$$\Delta x \Delta p \approx \frac{\hbar}{2}$$

$$\Delta p \approx \frac{\hbar}{2\Delta x} = \frac{1.05 \times 10^{-34}}{2 \times 50 \times 10^{-12}} = 1.05 \times 10^{-24} \text{ kg m/sec}$$

Problem 2: Prove that Heisenberg uncertainty principle is dimensionally correct?

Solution: Since from Heisenberg uncertainty principle we know that

$$\Delta x \Delta p \geq \frac{\hbar}{2} \dots \dots \dots (1)$$

Unit of Δx is meter $\Rightarrow [L]$

Unit of Δp is kg meter/sec $\Rightarrow [MLT^{-1}]$

So dimension of $\Delta x \Delta p = [ML^2T^{-1}]$

$$\text{Dimension of } \hbar = \frac{E}{\omega} = \frac{[ML^2T^{-2}]}{[T^{-1}]} = [ML^2T^{-1}]$$

Thus dimensions of equation (1) are balanced and this equation is dimensionally correct.

Problem 3: If the position of an electron in a membrane is measured to an accuracy of $1.00 \mu\text{m}$,

- (a) What is the electron's minimum uncertainty in momentum and velocity?
- (b) If the electron has this velocity, what is its kinetic energy in eV?

Solution: (a) Given that uncertainty in the position $\Delta x = 1 \mu\text{m} = 10^{-6} \text{ m}$

Now from Heisenberg uncertainty principle

$$\Delta x \Delta p \approx \frac{\hbar}{2}$$

$$\Delta p = m \Delta v \approx \frac{\hbar}{2\Delta x} = \frac{1.05 \times 10^{-34}}{2 \times 10^{-6}} = 5.25 \times 10^{-29} \text{ kg m/sec}$$

$$\Delta v \approx \frac{\hbar}{2 m \Delta x} = \frac{1.05 \times 10^{-34}}{2 \times 9.1 \times 10^{-31} \times 10^{-6}} = 57.6 \text{ m/sec}$$

(b) If electron velocity becomes $v = 57.6 \text{ m/sec}$

Then the kinetic energy of the electron

$$KE = \frac{m v^2}{2} = \frac{9.1 \times 10^{-31} \times (57.6)^2}{2} = 1.51 \times 10^{-27} \text{ Joules}$$

$$KE = \frac{1.51 \times 10^{-27}}{1.6 \times 10^{-19}} = 9.43 \times 10^{-9} \text{ eV} = 9.43 \text{ neV}$$

Problem 4: Suppose the velocity of an electron in an atom is known to an accuracy of $2 \times 10^3 \text{ m/sec}$. What is the electron's minimum uncertainty in position?

Solution: Given that uncertainty in the velocity of electron $\Delta v = 2 \times 10^3 \text{ m/sec}$

Therefore the uncertainty in momentum of the electron

$$\Delta p = m \Delta v = 9.1 \times 10^{-31} \times 2 \times 10^3 = 18.2 \times 10^{-28} \text{ kgm/sec}$$

$$\Delta x \approx \frac{\hbar}{2 \Delta p} = \frac{1.05 \times 10^{-34}}{2 \times 18.2 \times 10^{-28}} = 0.0288 \times 10^{-6} = 28.8 \approx 29 \text{ nm}$$

Problem 5: The velocity of a proton in an accelerator is known to an accuracy of 0.250% of the speed of light. What is the smallest possible uncertainty in its position?

Solution: Given that uncertainty in the velocity of electron $\Delta v = 0.25\% \text{ of the velocity of light } c$

$$\Delta v = 0.0025 \times 3 \times 10^8 \text{ m/sec}$$

Uncertainty in momentum

$$\Delta p = m \Delta v = 1.67 \times 10^{-27} \times 0.0025 \times 3 \times 10^8$$

$$\Delta p = 0.012525 \times 10^{-19} \text{ kgm/sec}$$

Uncertainty in position

$$\Delta x \approx \frac{\hbar}{2 \Delta p} = \frac{1.05 \times 10^{-34}}{2 \times 0.012525 \times 10^{-19}} = 4.19 \times 10^{-15}$$

Problem 6: Show that the particle which has uncertainty in position $\Delta x = \frac{\lambda}{2\pi}$ where λ is de-Broglie wavelength. Show that velocity uncertainty is same order of magnitude as the velocity itself.

Solution: Since from Heisenberg uncertainty principle we can write

$$\Delta x \Delta p \approx \frac{\hbar}{2} \dots \dots \dots (1)$$

Given that uncertainty in position $\Delta x = \frac{\lambda}{2\pi}$

Thus the uncertainty in velocity $\Delta v \approx \frac{\hbar}{2 m \Delta x} = \frac{\hbar \times 2\pi}{2 m \lambda} = \frac{h}{2m\lambda} \dots \dots \dots (1)$

From de-Broglie wavelength relation

$$\lambda = \frac{h}{mv}$$

$$v = \frac{h}{m \lambda} \dots \dots \dots (2)$$

Thus from equation (1) and (2) the velocity and uncertainty in velocity, are same order of magnitude.

1. Operators in Quantum mechanics

Physical Quantities	Operators	Symbol
Position	x	$\hat{X}$
Linear Momentum (x-component)	$-i\hbar \frac{d}{dx}$	$\hat{P}_x$
Kinetic Energy	$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$	$\hat{K}$
Potential Energy	$V(x)$	$\hat{V}$
Total Energy (free particle)	$i\hbar \frac{d}{dt}$	$\hat{E}$
Total Mechanical Energy (also known as Hamiltonian Operator)	$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$	$\hat{H}$

2. Expectation values of measurable quantities

(i) Expectation value of position:

Let us consider a particle is described by a wave function Ψ , the average value of the position x of the particle, is given by expectation value

$$\langle x \rangle = \frac{\int_{-\infty}^{\infty} \Psi^* x \Psi dx}{\int_{-\infty}^{\infty} \Psi^* \Psi dx} = \frac{\int_{-\infty}^{\infty} \Psi^* x \Psi dx}{\int_{-\infty}^{\infty} |\Psi|^2 dx}$$

If the function is normalized, i.e. $\int_{-\infty}^{\infty} \Psi^* \Psi dx = 1$

$$\text{Then } \langle x \rangle = \int_{-\infty}^{\infty} \Psi^* \hat{X} \Psi dx = \int_{-\infty}^{\infty} \Psi^* x \Psi dx = \int_{-\infty}^{\infty} x \Psi^* \Psi dx = \int_{-\infty}^{\infty} x |\Psi|^2 dx$$

(ii) Expectation value of momentum:

Expectation value of the momentum is defined by

$$\begin{aligned} \langle p \rangle &= \int_{-\infty}^{\infty} \Psi^* \hat{p} \Psi dx \\ \Rightarrow \langle p \rangle &= \int_{-\infty}^{\infty} \Psi^* \left(-i\hbar \frac{d}{dx} \right) \Psi dx \\ \Rightarrow \langle p \rangle &= -i\hbar \int_{-\infty}^{\infty} \Psi^* \left(\frac{d\Psi}{dx} \right) dx \end{aligned}$$

(iii) Expectation value of energy of the free particle:

Expectation value of the Energy is defined by

$$\begin{aligned} \langle E \rangle &= \int_{-\infty}^{\infty} \Psi^* \hat{E} \Psi dx \\ \Rightarrow \langle E \rangle &= \int_{-\infty}^{\infty} \Psi^* \left(i\hbar \frac{d}{dt} \right) \Psi dx \\ \Rightarrow \langle E \rangle &= i\hbar \int_{-\infty}^{\infty} \Psi^* \left(\frac{d\Psi}{dt} \right) dx \end{aligned}$$

(iv) Expectation value of kinetic energy of the particle:

Expectation value of the kinetic energy is defined by

$$\begin{aligned}\langle K \rangle &= \int_{-\infty}^{\infty} \Psi^* \hat{K} \Psi dx \\ \Rightarrow \langle K \rangle &= \int_{-\infty}^{\infty} \Psi^* \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \right) \Psi dx \\ \Rightarrow \langle K \rangle &= -\frac{\hbar^2}{2m} \int_{-\infty}^{\infty} \Psi^* \left(\frac{d^2 \Psi}{dx^2} \right) dx\end{aligned}$$

(v) Expectation value of potential energy of the particle:

Expectation value of the potential energy is defined by

$$\begin{aligned}\langle V \rangle &= \int_{-\infty}^{\infty} \Psi^* \hat{V} \Psi dx \\ \Rightarrow \langle V \rangle &= \int_{-\infty}^{\infty} \Psi^* V(x) \Psi dx\end{aligned}$$

3. Operators and Eigen Values:

Certain dynamical variables (*suppose* G) are restricted to a discrete values (G_n), or in other words G be quantized. If the system is defined by wave function Ψ_n is such that:

$$\hat{G} \Psi_n = G_n \Psi_n$$

Where, $\hat{G}$ is the operator corresponds to the physical quantity G , of which each discrete values G_n , is a real number.

Example: Eigen value of a total energy Operator (Hamiltonian Operator)

Suppose the Eigen values of a total energy operator $\hat{H}$ are given by discrete values (E_n), and if the system is defined by wave function Ψ_n is such that:

$$\begin{aligned}\hat{H} \Psi_n &= E_n \Psi_n \\ -\frac{\hbar^2}{2m} \frac{d^2 \Psi_n}{dx^2} + V(x) \Psi_n &= E_n \Psi_n \\ \frac{d^2 \Psi_n}{dx^2} + \frac{2m}{\hbar^2} (E_n - V(x)) \Psi_n &= 0\end{aligned}$$

After solving the above equation one can evaluate the Eigen values of the Hamiltonian operator.

4. Particle in a one dimensional infinite potential well:

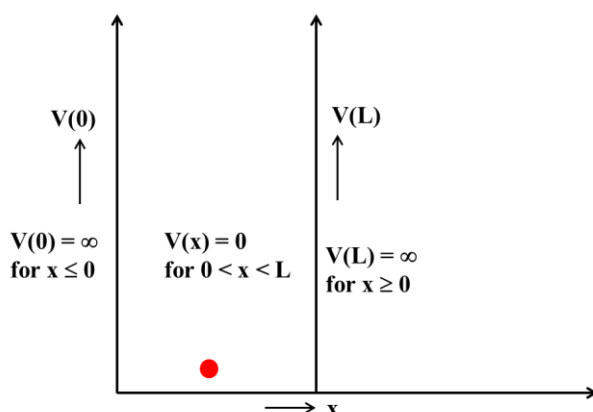


Fig. 1. A square potential well with infinitely high barriers at each end corresponds to a box with infinitely hard walls

Suppose a particle of mass m , is trapped in an infinite potential well whose boundaries are defined as $V(x) = 0$ for $0 < x < L$

$$V(x) = \infty \text{ for } x \leq 0 \text{ and } x \geq L$$

Please note that infinitely high potential well correspond infinitely hard walls which mean that particle is trapped in a one dimensional box having infinitely hard walls.

The trapped particle does not lose energy when it collides with the walls of the box.

We are supposed to find out the particle wave function Ψ , and energy Eigen values.

Suppose the particle's associated wave function is Ψ and energy Eigen values is E_n .

Now writing the time independent Schrodinger wave equation for the particle in the range $0 < x < L$.

$$\begin{aligned} \frac{d^2 \Psi_n}{dx^2} + \frac{2m}{\hbar^2} (E_n - V(x)) \Psi_n &= 0 \\ \Rightarrow \frac{d^2 \Psi_n}{dx^2} + \frac{2mE_n}{\hbar^2} \Psi_n &= 0 \dots\dots\dots(1) \end{aligned}$$

Suppose $k = \sqrt{\frac{2mE_n}{\hbar^2}}$ then equation (1) can be written as

$$\frac{d^2 \Psi_n}{dx^2} + k^2 \Psi_n = 0 \dots\dots\dots(2)$$

Equation (2) is the differential equation of second order and first degree. In order to solve this equation, first we will write the auxiliary equation of Equation (2) and find its roots.

Auxiliary equation of equation (2),

$$D^2 + k^2 = 0 \dots\dots\dots(3)$$

Note: In order to write auxiliary equation of any differential equation we use to represent $D \equiv \frac{d}{dx}$ and remove the wave function Ψ . Then we write rest of the quantities as shown in equation (3) and find the roots.

Roots of Equation (3), is $D = \pm ik$

Thus the solution of Equation (2) can be written as:

$$\Psi_n = A \sin kx + B \cos kx \dots \dots \dots (4)$$

Where A and B are constants and can be determined applying the boundary conditions. As you know the particle is strictly bounded in the region $0 < x < L$, so the particle wave function will be zero at $x = 0$ and L

Thus at $x = 0$ $\Psi_n = 0$ thus from Equation (4)

$$0 = A \sin 0 + B \cos 0 \Rightarrow B = 0$$

Thus the equation (4) can be written as

$$\Psi_n = A \sin kx \dots \dots \dots (5)$$

Again at $x = L$ $\Psi_n = 0$, therefore equation (5) gives

$$\Psi_n = A \sin kL = 0$$

$$\sin kL = 0$$

$$kL = n\pi \dots \dots \dots (6)$$

$$(kL)^2 = (n\pi)^2$$

$$\frac{2mE_n L^2}{\hbar^2} = n^2 \pi^2$$

$$E_n = \frac{\hbar^2 n^2 \pi^2}{2mL^2} \dots \dots \dots (7)$$

Equation (7) gives the energy Eigen values of the particle where $n = 1, 2, 3, \dots$

Substituting the value of $k = \frac{n\pi}{L}$ from equation (6) in equation (5) we have

$$\Psi_n = A \sin \frac{n\pi x}{L} \dots \dots \dots (8)$$

Applying normalization condition in order to find A

$$\int_{-\infty}^{\infty} \Psi_n^* \Psi_n dx = 1$$

$$\int_0^L A^2 \sin^2 \frac{n\pi x}{L} dx = 1$$

$$\int_0^L \frac{A^2}{2} 2 \sin^2 \frac{n\pi x}{L} dx = 1$$

$$\frac{A^2}{2} \int_0^L \left(1 - \cos \frac{2n\pi x}{L}\right) dx = 1$$

$$\frac{A^2}{2} \left[x - \frac{L}{2n\pi} \sin \frac{2n\pi x}{L} \right]_0^L = 1$$

$$A = \sqrt{\frac{2}{L}}$$

Thus equation (5) can be written as

$$\Psi_n = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \dots \dots \dots (9)$$

Equation (9) is the Eigen function of the particle where n = 1,2,3,.....

Let us plot the wave function and probability density of the particle for different n values

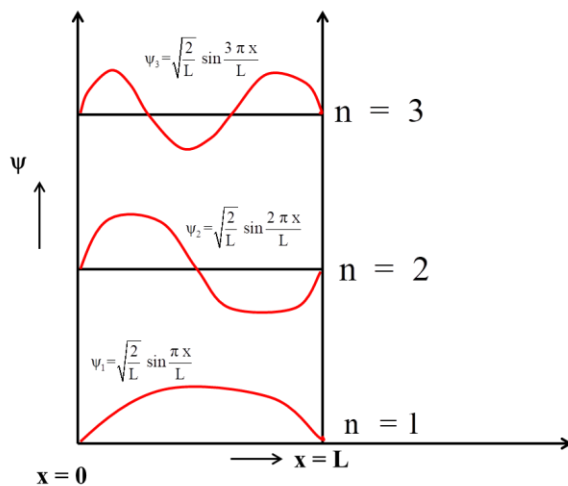


Fig. 2. Wave function plot

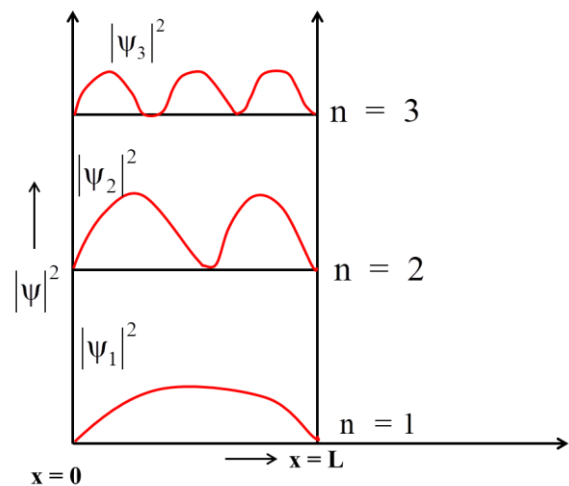


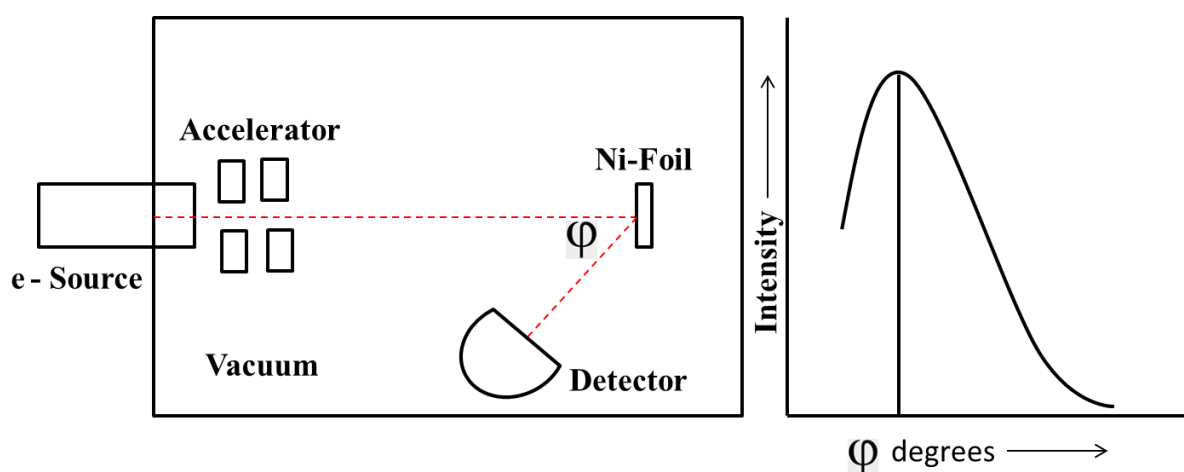
Fig. 3. Probability density of a particle

Wave nature of particles

- (i) *An experiment that confirms the de-Broglie hypothesis.*
- (ii) *An experiment which shows the wave nature of particles.*

In 1927 Clinton Davisson and Lester Germer in US and G. P. Thomson in England, independently showed the particle diffraction from the regular atomic array of crystals. This experiment showed the wave nature of the particle as they diffracted from the atomic arrays of crystals, and the diffraction is related to wave properties. Thus particles also shows the wave nature, which is exactly de-Broglie hypothesis says.

Davisson and Germer Experiment:



In 1927 Davisson and Germer, doing an experiment on angular distribution of electrons scattered from the Ni- foil. The experimental arrangement is shown in the above figure. They had no intention of testing de-Broglie hypothesis of wave character of material particles. But due to some unexpected problems, they have found air leakage in the vacuum chamber in which the Ni- foil was placed. Due to air contact the Ni-foil were oxidized, and in order to perform, their aimed experiment, they have given the heat treatment to expel the oxygen from the Ni-foil. After this treatment, they found that the bombarded electrons were scattered intensely at a specific angle ϕ , which depended on the initial KE of the bombarded electrons. This result was similar to the x-ray diffraction patters. **Davisson and**

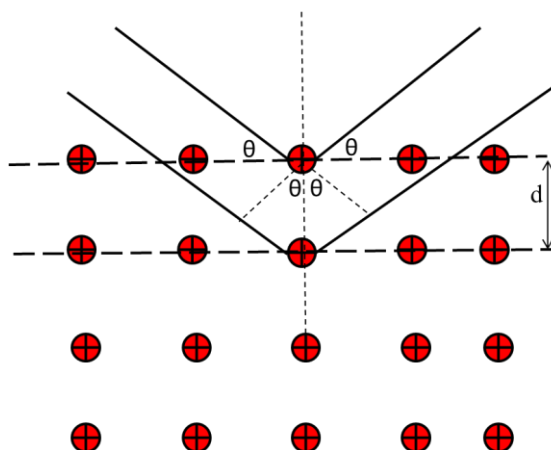
Germer concluded that they were observing the wave character of the electrons as proposed by de-Broglie.

Question: Why did the result changes (in Davisson and Germer Experiment) when the Ni-foil was given heat treatment?

Answer: After the heat treatment Nickel got crystallized, and the Ni-atoms got arranged in a regular fashion. Such an ordered array of atoms reflects the x-rays in a very specific direction. This is a common phenomenon and this fact is used to determine the crystal structure using x-rays.

Question: How x-rays diffracted /reflected only in a specific directions, are related to the crystal structure?

X-ray Diffraction:



Answer: Suppose x-rays are sent to a crystal making an angle θ , with the plane of the atoms (as shown in above figure). A part of x-ray waves were diffracted from the top plane of the atoms, and another part are from next plane and so on. The difference in path length travelled by the waves diffracted from the consecutive planes $2d \sin \theta$. Here d , is the distance between the **two consecutive planes**. **The diffracted wave has large intensity for directions satisfying the conditions**

$$2d \sin \theta = n\lambda \dots \dots \dots (1)$$

Where λ , is the incident x-ray photon wavelength, n is integer 1,2,3,4.....

Question: How Davisson and Germer, came to the conclusion that the wave associated with the electrons was diffracted from the planes of the atoms (Ni-atoms) in the same way as x-ray photons diffracted from the atomic planes?

Answer: Let us suppose the incident electron beam falling on the Ni-Foil normal to its surface (in Davisson Germer experiment) having kinetic energy ($KE = 54 \text{ eV}$), and scattered strongly at an angle $\phi = 50^\circ$ with the normal to the surface. The incidence angle of the electron beam from the surface responsible for the scattering $\theta = 65^\circ$. Suppose the separation between two consecutive planes $d = 0.909 \text{ nm}$. The wavelength of the electron waves calculated using

$$2d \sin \theta = \lambda$$

Here $n = 1$ is taken as the first order diffraction.

$$\lambda = 2d \sin \theta = 2 \times 0.909 \sin 65$$

$$\lambda = 0.165 \text{ nm}$$

This is the wavelength of the wave associated with the electron as obtained from the Davisson & Germer experiment.

Now if the wavelength of the associated wave of an electron calculated from the de-Broglie hypothesis is same as calculated above, then it will be prove the de-Broglie Hypothesis.

$$\lambda = \frac{h}{p}$$

Momentum of the incident electrons

$$p = \sqrt{2mK.E.} = \sqrt{2 \times 9.1 \times 10^{-31} \times 54 \times 1.6 \times 10^{-19}}$$

$$p = 3.965 \times 10^{-24} \text{ kgm/sec}$$

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34}}{3.965 \times 10^{-24}} = 0.167 \text{ nm}$$

Thus the result from Davisson & Germer experiment agrees extremely well with the de-Broglie Hypothesis.

1. Wave function Ψ :

The quantity whose variation make up matter wave is called wave function Ψ .

2. **Matter Wave:** The wave associated with the materialistic particles and represented by Ψ is called the matter wave.

$$\Psi = \Psi(x, y, z, t) \dots \dots \dots (1)$$

The wave function Ψ is dependent upon the position and time as given in equation no (1). Equation (1) represents the value of the wave function associated with moving particle at **position (x,y,z,t)**.

3. Born Interpretation:

The wave function Ψ is not an observable quantity, because the wave function represents the matter wave associated with the particle like electron. Further if we consider an electromagnetic wave or sound waves, the intensity at any point can simply be obtained by squaring the amplitude i.e. $I \propto E_0^2$ where E_0 the amplitude of the electromagnetic wave represented by $E = E_0 e^{-i(\omega t - kx)}$.

In the same way matter wave associated with material particles like electron, represented by wave function Ψ , the square of the amplitude of matter wave (Ψ^2) gives the **probability of finding the particle**. This interpretation was first made by Max Born and called the Born interpretation.

4. Probability of finding the particle:

Probability of finding the particle which is described by wave function Ψ at the point (x,y,z) and at time t is proportional to the value $|\Psi|^2$.

Usually the wave function Ψ is represented by

$$\Psi = A e^{-i(\omega t - kx)} \dots \dots \dots (2)$$

The wave function ψ is usually a complex quantity having real and imaginary parts. $\left(\omega = \frac{E}{\hbar} \text{ sec}^{-1}\right)$ is the angular frequency of the associated wave and $\left(k = \frac{p}{\hbar} \text{ m}^{-1}\right)$ is the wavenumber. E is the energy and p is the momentum of the associated wave with the particle.

$$\text{So the probability of finding the particle is } P = |\Psi|^2 \dots \dots \dots (3)$$

Probability of finding the particle in the range (x_1 to x_2) can be written as

$$P_{x_1, x_2} = \int_{x_1}^{x_2} |\Psi|^2 dx = \int_{x_1}^{x_2} \Psi \Psi^* dx \dots \dots \dots (4)$$

Where $\Psi^* = A e^{i(\omega t - kx)}$ complex conjugate of the wave function

$$\Psi = A e^{-i(\omega t - kx)} \text{ represented by equation (2)}$$

5. Normalization:

Suppose wave function ψ is represented by

$$\Psi = A e^{-i(\omega t - kx)} \dots \dots \dots (5)$$

complex conjugate of the wave function represented by equation (5)

$$\Psi^* = A e^{i(\omega t - kx)} \dots \dots \dots (6)$$

Therefore a wave function is said to be normalized if

$$\int_{-\infty}^{\infty} |\Psi|^2 dx = \int_{-\infty}^{\infty} \Psi \Psi^* dx = 1 \dots \dots \dots (7)$$

From equation (3) equation (7) can be written as

$$\text{or } \int_{-\infty}^{\infty} P dx = 1 \dots \dots \dots (8)$$

Thus Equation (7) or (8) represents the normalization condition.

6. Properties of wave function Ψ :

- (a) Ψ must be continuous everywhere.
- (b) Ψ must be single valued
- (c) $\frac{\partial \Psi}{\partial x}$, $\frac{\partial \Psi}{\partial y}$, and $\frac{\partial \Psi}{\partial z}$ must be continuous and single values everywhere.
- (d) ψ must be normalizable which means that
 $\Psi \rightarrow 0$ and $x \rightarrow \pm\infty$, $y \rightarrow \pm\infty$ and $z \rightarrow \pm\infty$ in order that
 $\int_{-\infty}^{\infty} |\Psi|^2 dx = \int_{-\infty}^{\infty} \Psi \Psi^* dx = 1$

7. Schrodinger's Equation:

Schrodinger's equation is fundamental principle of Quantum Mechanics: this cannot be derived from any other basic principles of physics.

(a) Time dependent Schrodinger Equation:

Suppose a wave function Ψ associated with a particle moving freely in $+x$, direction is given by

$$\Psi = Ae^{-i(\omega t - kx)} \dots \dots \dots (1)$$

Where $\left(\omega = \frac{E}{\hbar} \text{ sec}^{-1}\right)$ is the angular frequency and $\left(k = \frac{p}{\hbar} \text{ m}^{-1}\right)$ is the wave number of the associated wave with the particle.

So wave function can be written as, after replacing the angular frequency and wavenumber

$$\Psi = Ae^{-i\left(\frac{Et}{\hbar} - \frac{px}{\hbar}\right)}$$

$$\Psi = Ae^{-\frac{i}{\hbar}(Et - px)} \dots \dots \dots (2)$$

Differentiating Equation (2) twice with respect to x , we have

$$\frac{d\Psi}{dx} = A \frac{ip}{\hbar} e^{-\frac{i}{\hbar}(Et - px)}$$

$$\frac{d^2\Psi}{dx^2} = A \left(\frac{ip}{\hbar}\right)^2 e^{-\frac{i}{\hbar}(Et - px)}$$

$$\frac{d^2\Psi}{dx^2} = -\left(\frac{p}{\hbar}\right)^2 A e^{-\frac{i}{\hbar}(Et - px)}$$

$$\frac{d^2\Psi}{dx^2} = -\frac{p^2\Psi}{\hbar^2} \dots\dots\dots(3)$$

$$\Rightarrow p^2\Psi = -\hbar^2 \frac{d^2\Psi}{dx^2} \dots\dots\dots(4)$$

Again differentiating equation (2) with respect to t , we have

$$\frac{d\Psi}{dt} = A \left(-\frac{iE}{\hbar} \right) e^{-\frac{i}{\hbar}(Et-px)}$$

$$\frac{d\Psi}{dt} = \left(-\frac{iE\Psi}{\hbar} \right) \dots\dots\dots(5)$$

$$\Rightarrow E\Psi = i\hbar \frac{d\Psi}{dt} \dots\dots\dots(6)$$

Total energy E , of the moving particle can be written as,

Total energy of the particle = Kinetic energy + Potential energy

$$E = \frac{p^2}{2m} + V(x, t) \dots\dots\dots(7)$$

Multiplying both the sides by ψ in equation (7), we have

$$E\Psi = \frac{p^2\Psi}{2m} + V(x, t)\Psi \dots\dots\dots(8)$$

Now substituting the value of $E\psi$ and $p^2\psi$ in equation (8) we have

$$i\hbar \frac{d\Psi}{dt} = -\frac{\hbar^2}{2m} \frac{d^2\Psi}{dx^2} + V\Psi \dots\dots\dots (9)$$

Equation (9) represents the time dependent Schrodinger Equation

For three dimensions, Equation (9) can be written as

$$i\hbar \frac{d\Psi}{dt} = -\frac{\hbar^2}{2m} \left(\frac{\partial^2\Psi}{\partial x^2} + \frac{\partial^2\Psi}{\partial y^2} + \frac{\partial^2\Psi}{\partial z^2} \right) + V\Psi \dots\dots (10)$$

(b) Time independent Schrodinger Equation or (Steady state form of the Schrodinger Equation)

In many situations the potential energy of the particle V , does not depend on time. The forces which acts on it and hence the potential V , vary with position only.

In this case Schrodinger equation can be simplified by removing the time t dependence.

So the wave function Ψ can be written as:

$$\Psi = Ae^{-\frac{i}{\hbar}(Et - px)} = Ae^{-i\frac{Et}{\hbar}} e^{i\frac{px}{\hbar}}$$

$$\Psi = \psi e^{-i\frac{Et}{\hbar}} \dots\dots\dots(1)$$

Where Ψ is the product of time dependent function, $e^{-i\frac{Et}{\hbar}}$, and position dependent function $\psi = Ae^{i\frac{px}{\hbar}}$.

We have already discussed the time dependent form of the Schrodinger equation

$$i\hbar \frac{d\Psi}{dt} = -\frac{\hbar^2}{2m} \frac{d^2\Psi}{dx^2} + V\Psi \dots\dots\dots(2)$$

Substituting the value of equation (1) in (2) we have

$$i\hbar \frac{d\psi e^{-i\frac{Et}{\hbar}}}{dt} = -\frac{\hbar^2}{2m} \frac{d^2\psi e^{-i\frac{Et}{\hbar}}}{dx^2} + V\psi e^{-i\frac{Et}{\hbar}}$$

$$\Rightarrow i\hbar \psi \frac{d e^{-i\frac{Et}{\hbar}}}{dt} = -\frac{\hbar^2}{2m} e^{-i\frac{Et}{\hbar}} \frac{d^2\psi}{dx^2} + V\psi e^{-i\frac{Et}{\hbar}}$$

$$\Rightarrow i\hbar \left(-\frac{iE}{\hbar}\right) \psi e^{-i\frac{Et}{\hbar}} = -\frac{\hbar^2}{2m} e^{-i\frac{Et}{\hbar}} \frac{d^2\psi}{dx^2} + V\psi e^{-i\frac{Et}{\hbar}}$$

$$\Rightarrow E \psi = -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi$$

$$\Rightarrow \frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V)\psi = 0 \dots\dots\dots (3)$$

Equation (3) is known as time independent Schrodinger Equation for one dimension.

For three dimensions, Equation (3) can be written as

$$\left(\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} \right) + \frac{2m}{\hbar^2} (E - V) \Psi = 0 \dots\dots\dots(4)$$

1. Particle in a one dimensional finite potential well:

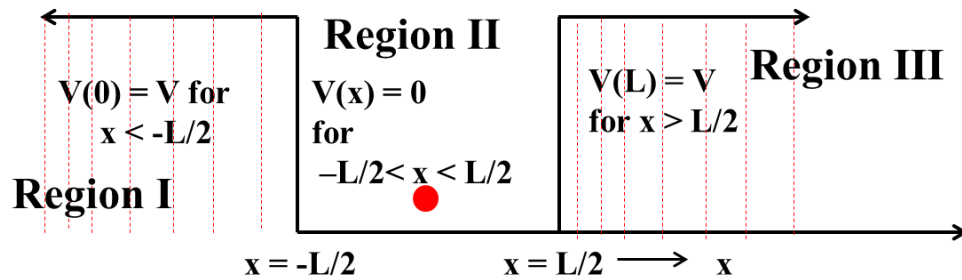


Fig. 1. A square potential well with finite height barriers at each end.

Previously discussed case of infinite potential well, is not possible in real world as the potential energies cannot be infinite. However the potential well with finite barrier does exist. Let us discuss the energy Eigen values and Eigen functions if particle is trapped in finite potential well.

Suppose a particle of mass m , is trapped in a symmetric finite potential well whose boundaries are defined as $V(x) = 0$ for $-\frac{L}{2} < x < \frac{L}{2}$

$$V(x) = V \text{ for } x < -\frac{L}{2} \text{ and } x > \frac{L}{2}$$

Suppose the particle energy $E < V$ (height of the barrier). Classically the particle with energy E will be bounces back and forth in the finite potential well without entering the region I and III (refer to Fig. 1). But quantum mechanics says there is certain probability of penetration in the region I and III even though $E < V$.

(i) Writing the Schrodinger (Time independent) equation for Region I

Suppose the particle wave function in region I is Ψ_I

$$\frac{d^2\Psi_I}{dx^2} + \frac{2m}{\hbar^2} (E - V)\Psi_I = 0 \dots\dots\dots(1)$$

Suppose $k_1 = \sqrt{\frac{2m(V-E)}{\hbar^2}}$ then equation (1) can be written as

$$\frac{d^2\Psi_I}{dx^2} - k_1^2\Psi_I = 0 \dots\dots\dots(2)$$

Equation (2) is the differential equation of second order and first degree. In order to solve this equation, first we will write the auxiliary equation of Equation (2) and find its roots.

Auxiliary equation of equation (2),

$$D^2 - k_1^2 = 0 \dots\dots\dots (3)$$

Note: In order to write auxiliary equation of any differential equation we use to represent $D \equiv \frac{d}{dx}$ and remove the wave function. Then we write rest of the quantities as shown in equation (3) and find the roots.

Roots of Equation (3), is $D = \pm k_1$

Thus the solution of Equation (2) can be written as:

$$\Psi_I = Ae^{k_1x} + Be^{-k_1x} \dots\dots\dots (4)$$

Since property of wave function of the particle says that

Ψ_I must be finite everywhere and $\Psi_I \rightarrow 0$ as $x \rightarrow -\infty$ for region I

Since equation (4) has two terms first term Ae^{k_1x} satisfy both the properties of wave function that is $\Psi_I \rightarrow 0$ as $x \rightarrow -\infty$ and Ψ_I is finite everywhere. But second term of equation (4), Be^{-k_1x} does not satisfy above conditions and wave function $\Psi_I \rightarrow \infty$ as $x \rightarrow -\infty$ so this term is useless and has to be dropped.

Thus the final solution of equation (2)

$$\Psi_I = Ae^{k_1x} \dots\dots\dots (5)$$

(ii) Writing the Schrodinger (Time independent) equation for **Region II**

$$\frac{d^2\Psi_{II}}{dx^2} + \frac{2mE}{\hbar^2}\Psi_{II} = 0 \dots\dots\dots (6)$$

Suppose $k_2 = \sqrt{\frac{2mE}{\hbar^2}}$ then equation (6) can be written as

$$\frac{d^2\Psi_{II}}{dx^2} + k_2^2\Psi_{II} = 0 \dots\dots\dots (7)$$

Solution of equation(7)

$$\Psi_{II} = C \sin k_2x + D \cos k_2x$$

(iii) Writing the Schrodinger (Time independent) equation for **Region III**

$$\frac{d^2\Psi_{III}}{dx^2} - k_1^2\Psi_{III} = 0 \dots\dots\dots (8)$$

Solution of equation (10) can be written as

$$\Psi_{III} = Ee^{-k_1x} \dots\dots\dots (9)$$

There are two types of wave functions obtained in :solutions even and odd:

Case I: Even functions

$$\Psi_I = Ae^{k_1x} \text{ for } x < -\frac{L}{2} \dots\dots\dots (10)$$

$$\Psi_{II} = D \cos k_2x \text{ for } -\frac{L}{2} < x < \frac{L}{2} \dots\dots\dots (11)$$

$$\Psi_{III} = Ae^{-k_1x} \text{ for } x > \frac{L}{2} \dots\dots\dots (12)$$

As wave function is even so the coefficients in I and third regions will be same.

Which means $A = E$

Applying the boundary conditions at $x = -\frac{L}{2}$, as wave function must be continuous at this point.

$$\Psi_I = \Psi_{II}$$

$$Ae^{-\frac{k_1L}{2}} = D \cos \frac{k_2L}{2} \dots\dots\dots (13)$$

Further first derivative of the wave function must be continuous at $x = -\frac{L}{2}$

$$\frac{d\Psi_I}{dx} = \frac{d\Psi_{II}}{dx}$$

$$Ak_1e^{-\frac{k_1L}{2}} = D k_2 \cos \frac{k_2L}{2} \dots\dots\dots (14)$$

Dividing Eq. (13) by (12) we have

$$\tan \frac{k_2L}{2} = \frac{k_1}{k_2} \dots\dots\dots (14)$$

For mathematical simplicity we write

$$\xi = \frac{k_2L}{2} \quad \text{and} \quad \eta = \frac{k_1L}{2}$$

Then we can write $\xi^2 + \eta^2 = \frac{L^2}{4} [k_2^2 + k_1^2] = \frac{L^2}{4} \left[\frac{2m(V-E)}{\hbar^2} + \frac{2mE}{\hbar^2} \right] = \frac{L^2}{4} \frac{2mV}{\hbar^2} = R^2 \dots\dots (15)$

Therefore equation (14) can be written as

$$\eta = \xi \tan \xi \dots\dots\dots (16)$$

Now substituting the value of η , from equation (15), equation (16) can be written as.

$$\sqrt{R^2 - \xi^2} = \xi \tan \xi \dots \dots \dots (17)$$

This is the solution of even function.

Case I: odd functions

$$\Psi_I = Ae^{k_1 x} \text{ for } x < -\frac{L}{2} \dots \dots \dots (18)$$

$$\Psi_{II} = C \sin k_2 x \text{ for } -\frac{L}{2} < x < \frac{L}{2} \dots \dots \dots (19)$$

$$\Psi_{III} = -Ae^{-k_1 x} \text{ for } x > \frac{L}{2} \dots \dots \dots (20)$$

As wave function is odd, so the coefficients in I and third regions will be same but opposite in sign.

Which means $A = -E$

Applying the boundary conditions at $x = -\frac{L}{2}$, as wave function must be continuous at this point.

$$\Psi_I = \Psi_{II}$$

$$Ae^{-\frac{k_1 L}{2}} = -C \sin \frac{k_2 L}{2} \dots \dots \dots (21)$$

Further first derivative of the wave function must be continuous at $x = -\frac{L}{2}$

$$\frac{d\Psi_I}{dx} = \frac{d\Psi_{II}}{dx}$$

$$Ak_1 e^{-\frac{k_1 L}{2}} = C k_2 \cos \frac{k_2 L}{2} \dots \dots \dots (22)$$

Dividing Eq. (22) by (21) we have

$$-\cot \frac{k_2 L}{2} = \frac{k_1}{k_2} \dots \dots \dots (23)$$

For mathematical simplicity we write

$$\xi = \frac{k_2 L}{2} \quad \text{and} \quad \eta = \frac{k_1 L}{2}$$

Then we can write $\xi^2 + \eta^2 = \frac{L^2}{4} [k_2^2 + k_1^2] = \frac{L^2}{4} \left[\frac{2m(V-E)}{\hbar^2} + \frac{2mE}{\hbar^2} \right] = \frac{L^2}{4} \frac{2mV}{\hbar^2} = R^2 \dots \dots \dots (24)$

Therefore equation (14) can be written as

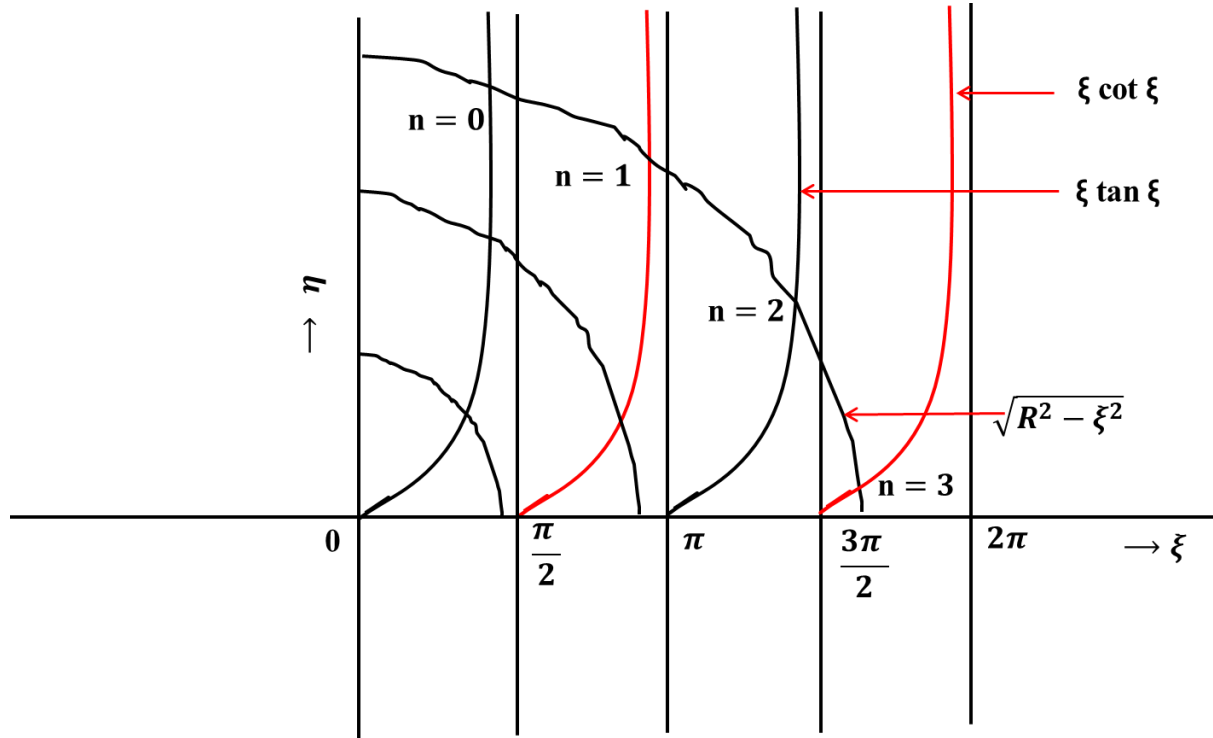
$$\eta = \xi \tan \xi \dots \dots \dots (26)$$

Now substituting the value of η , from equation (15), equation (16) can be written as.

$$\sqrt{R^2 - \xi^2} = -\xi \cot \xi \dots \dots \dots (27)$$

This is the solution of odd function.

The solutions of equations (17) and (27) form a discrete set. As left side of equations (17) and (27) represents a circle and right hand side represents trigonometric functions. Therefore the intersection of the circle and trigonometric curve yield the solution.



The number of solutions depends upon the size of radius R , which in turn depends upon the depth V , and width L , of the potential well.

(i) Case I

When $0 < R < \frac{\pi}{2}$

$$0 < R^2 < \left(\frac{\pi}{2}\right)^2$$

$$0 < \frac{mVL^2}{2\hbar^2} < \left(\frac{\pi}{2}\right)^2$$

$$0 < V < \left(\frac{\pi}{2}\right)^2 \frac{2\hbar^2}{mL^2}$$

There is only one bound state corresponding to $n=0$

(ii) Case II

When $\frac{\pi}{2} < R < \pi$

$$\left(\frac{\pi}{2}\right)^2 < R^2 < \pi^2$$

$$\left(\frac{\pi}{2}\right)^2 < \frac{mVL^2}{2\hbar^2} < \pi^2$$

$$\left(\frac{\pi}{2}\right)^2 \frac{2\hbar^2}{mL^2} < V < \pi^2 \frac{2\hbar^2}{mL^2}$$

There are two bound states $n=0$ (corresponding to even function) and $n=1$ (corresponding to odd function)

(iii) Case III

$$\text{When } \pi < R < \frac{\pi}{2}$$

$$\pi^2 < R^2 < \left(\frac{3\pi}{2}\right)^2$$

$$\pi^2 \frac{2\hbar^2}{mL^2} < V < \left(\frac{3\pi}{2}\right)^2 \frac{2\hbar^2}{mL^2}$$

There are three bound states $n=0$ and 2 (corresponding to even function) and $n=1$ (corresponding to odd function)

In general well width at which n states are allowed is given by

$$R = \frac{n\pi}{2}$$

$$R^2 = \left(\frac{n\pi}{2}\right)^2$$

$$V = \left(\frac{\pi}{2}\right)^2 \frac{2\hbar^2}{mL^2} n^2$$

Therefore the system consists of set of alternate even and odd states, lowest state called ground state is even state, next state called first excited state is odd state, and so on.

In the limiting case

For $V \rightarrow \infty$ the LHS of equation (17) will become

$$\tan \xi = \tan \frac{k_2 L}{2} = \infty = \tan \left(n + \frac{1}{2}\right) \pi$$

$$\Rightarrow \frac{k_2 L}{2} = \left(n + \frac{1}{2}\right) \pi$$

$$\Rightarrow \left(\frac{k_2 L}{2}\right)^2 = \left(n + \frac{1}{2}\right)^2 \pi^2$$

$$\Rightarrow \frac{2mEL^2}{4\hbar^2} = \left(n + \frac{1}{2}\right)^2 \pi^2$$

$$\Rightarrow E = \frac{4\left(n + \frac{1}{2}\right)^2 \pi^2 \hbar^2}{2mL^2}$$

$$\Rightarrow E = \frac{(2n+1)^2 \pi^2 \hbar^2}{2mL^2} \quad \text{Where } n = 0, 1, 2, 3, 4, \dots$$

Also for $V \rightarrow \infty$ the LHS of equation (27) will become

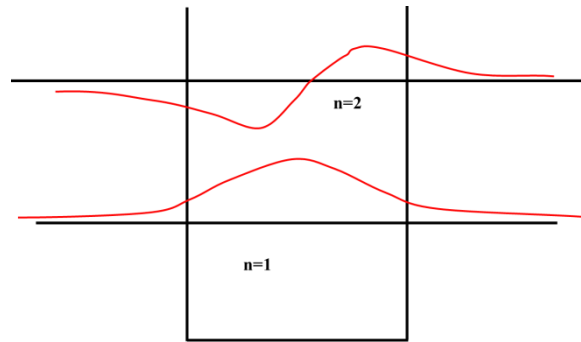
$$\cot \xi = \cot \frac{k_2 L}{2} = \infty = \cot n\pi$$

$$\Rightarrow \frac{k_2 L}{2} = n\pi$$

$$\Rightarrow \left(\frac{k_2 L}{2} \right)^2 = (n\pi)^2$$

$$\Rightarrow \frac{2mEL^2}{4\hbar^2} = (n\pi)^2$$

$$\Rightarrow E = \frac{(2n\pi)^2 \hbar^2}{2mL^2} \text{ Where } n = 1, 2, 3, 4, \dots$$



Wave function of the particle in finite symmetric potential well